

## 11. For the above application create authorized end points using JWT (JSON Web Token)

### **Install**

```
npm install express jsonwebtoken
```

### **app.js**

```
const express = require('express');
const jwt = require('jsonwebtoken');
const app = express();
app.use(express.json());

const SECRET = "key123";
let students = [];

// LOGIN → Generate Token
app.post('/login', (req,res)=>{
  const token = jwt.sign({user:"admin"}, SECRET);
  res.json({token});
});

// Middleware (Protect API)
const verify = (req,res,next)=>{
  let token = req.headers['authorization'];
  if(!token) return res.send("No Token");

  jwt.verify(token, SECRET, (err,data)=>{
    if(err) return res.send("Invalid Token");
    next();
  });
};

// CRUD (Protected)
app.post('/add', verify, (req,res)=>{
  students.push(req.body);
  res.json({msg:"Added", data:students});
});

app.get('/all', verify, (req,res)=>{
  res.json(students);
});

app.put('/update/:id', verify, (req,res)=>{
  students[req.params.id]=req.body;
```

```
app.delete('/delete/:id', verify, (req,res)=>{
    students.splice(req.params.id,1);
    res.json({msg:"Deleted", data:students});
});
```

outputs:

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

"No Token"

Authorization: eyJhbGciOiJIUzI1Ni...

[ ]

```
{
  "msg": "Added",
  "data": [
    { "name": "ram" }
  ]
}
```

